L13-14: Quantum Computing Primitives

Vishvendra S. Poonia
Department of Electronics and Communication Engineering
Indian Institute of Technology Roorkee
vishvendra@ece.iitr.ac.in
vishvendra.punia@gmail.com

This session

Density Matrix

Coherence

Purity

Entanglement

Qubits: Bloch Sphere

Mathematically, a qubit lives in a 2D complex Hilbert space: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

with normalization: $|\alpha|^2 + |\beta|^2 = 1$

For a general qubit state: $\alpha, \beta \in C \Rightarrow$ **4 real parameters**

- Normalization removes one parameter.
- Two states differing only by a global phase are physically identical: $|\psi\rangle \sim e^{i\gamma}|\psi\rangle$. That removes one more parameter.
- The physical qubit state has 4-2 = 2 parameters
- **A qubit has 2 real degrees of freedom.**
- **This is exactly the surface of a sphere**

Amplitude in the polar form:

$$\alpha = |\alpha| e^{ia}, \beta = |\beta| e^{ib}$$

By factoring out e^{ia}**:** $|\psi\rangle = e^{ia}(|\alpha||0\rangle + |\beta| e^{i(b-a)}|1\rangle)$

Drop the global phase e^{ia}**:**

$$|\psi\rangle = |\alpha||0\rangle + |\beta| e^{i(b-a)}|1\rangle$$

So, we have: $|\psi\rangle = |\alpha||0\rangle + |\beta| e^{i\phi}|1\rangle$**;** $\phi = b - a$

From the normalization: $|\alpha|^2 + |\beta|^2 = 1$**,**

Therefore, we can parametrize: $|\alpha| = cos(\theta/2)$, $|\beta| = sin(\theta/2)$ for some $\theta \in [0, \pi]$.where: $0 \le \theta \le \pi$, and $0 \le \phi < 2\pi$

Final canonical form: $|\psi\rangle = cos(\theta/2)|0\rangle + e^{i\phi} sin(\theta/2)|1\rangle$**; where:** $0 \le \theta \le \pi$**, and** $0 \le \phi < 2\pi$

These are spherical coordinates.

Mapping: $r = (sin\theta cos\phi, sin\theta sin\phi, cos\theta)$

Each pure state corresponds to a point on the surface of the unit sphere in $\mathbb{R}^3$**.**

Qubits: Bloch Sphere

Mathematically, a qubit lives in a 2D complex Hilbert space: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$

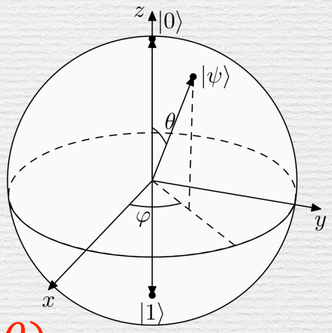
with normalization: $|\alpha|^2 + |\beta|^2 = 1$

Removing Redundancy: Global Phase

- Two states differing only by a global phase are physically identical: $|\psi\rangle \sim e^{i\gamma}\,|\psi\rangle$
- **After removing:**
  - **N**ormalization (1 constraint)
  - Global phase (1 constraint)
- **A qubit has 2 real degrees of freedom.**
- **This is exactly the surface of a sphere**

Every pure qubit state can be written as:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

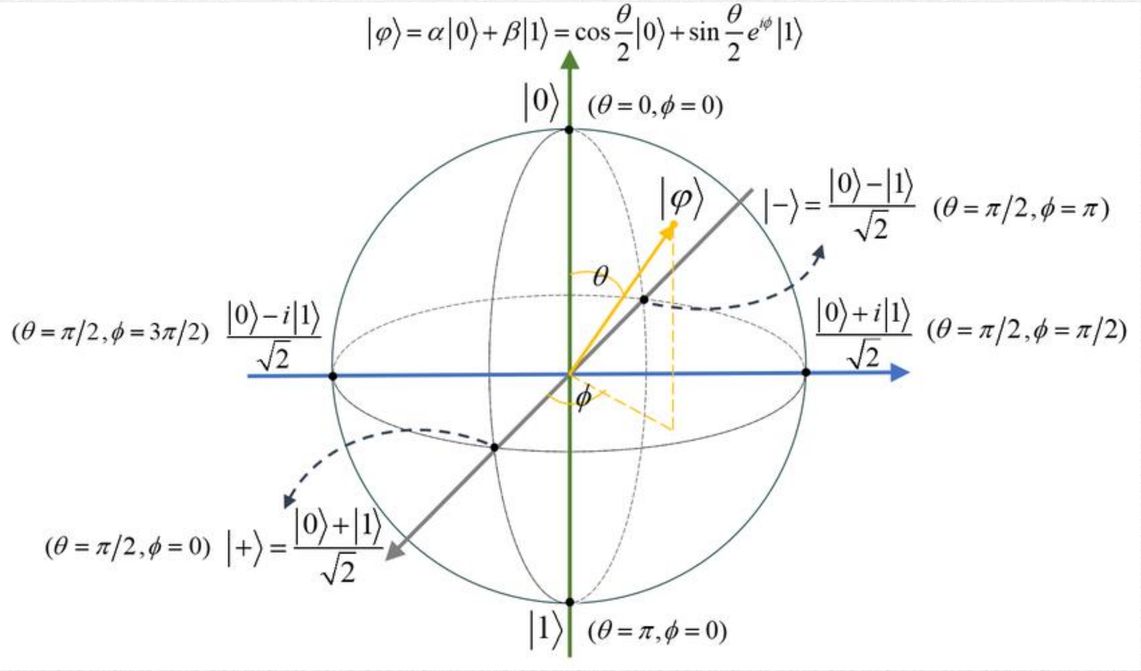


where: $0 \le \theta \le \pi$**, and** $0 \le \phi < 2\pi$

These are spherical coordinates.

Mapping: $r = (sin\theta cos\phi, sin\theta sin\phi, cos\theta)$

Each pure state corresponds to a point on the surface of the unit sphere in $\mathbb{R}^3$.



Ref: DOI:10.1109/TC.2025.3557968

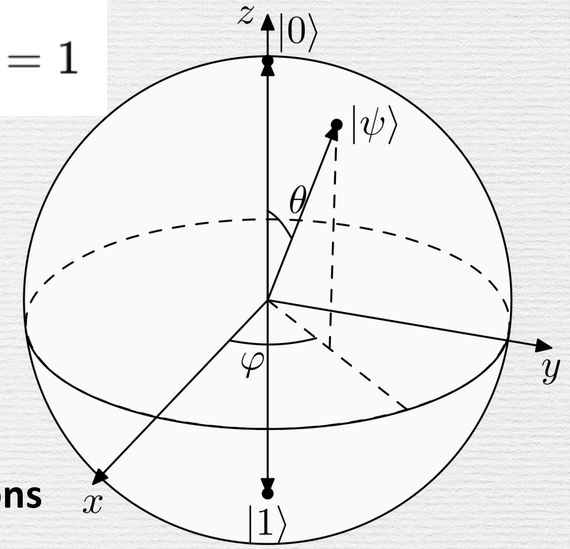
The Bloch Sphere

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

A qubit state ↔ point on the surface of a unit sphere

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

$$(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$$



Various Parameters:

- **θ (polar angle)** → controls population between $|0\rangle$ and $|1\rangle$
- **ϕ (azimuthal angle)** → controls *relative phase*

Axes interpretation:

- **+Z** → $|0\rangle$
- **−Z** → $|1\rangle$
- **+X** → $(|0\rangle + |1\rangle)/\sqrt{2}$
- **+Y** → $(|0\rangle + i|1\rangle)/\sqrt{2}$

Global phase is invisible — only relative phase matters (also evident via Bloch sphere representation)

Converts complex amplitudes into angles and directions

Helps visualize: Superposition, Phase differences, Orthogonality (antipodal points)

Quantum Gates = Rotations

- Single-qubit gates become **physical rotations**
  - **X gate → 180° rotation about X-axis**
  - **Hadamard → rotation placing state on equator**
  - **Phase gate → rotation around Z-axis**
- It turns abstract matrix math into **spatial motion**
- **Mixed states → inside the sphere**
- Decoherence → shrinking Bloch vector
- T_1 and T_2 relaxation processes are easily visualized
- **Measurement in Z-basis:**
  - Probability of $|0\rangle = \cos^2(\theta/2)$
  - Probability of $|1\rangle = \sin^2(\theta/2)$

Collapse = **projection onto an axis**

Measurement is geometric now!

Density Matrix

Why Do We Need Density Matrices?

- So far, we described quantum states as: $|\psi\rangle \in H$
- This works perfectly for:
  - Isolated systems
  - Pure states
  - Fully known quantum states
- But real quantum systems:
  - Interact with environment
  - Become entangled with other systems
  - May be classically uncertain mixtures
  - Are often partially observed
- The state vector formalism is not enough.
- We need a more general description. That description is the density operator (density matrix).

From Ket to Density Operator

- **Pure State Case: Given a normalized state $|\psi\rangle$**
  - **We define the density operator: $\rho = |\psi\rangle\langle\psi|$**
  - **This is the outer product of the ket and bra.**

Key Points:

- **Every pure state has a density matrix.**
- **But not every density matrix corresponds to a pure state.**
- **Here comes the notion of Mixed states.**

Density Matrix

Pure State: Given a normalized state $|\psi\rangle$

- We define the density operator: $\rho = |\psi\rangle\langle\psi|$

Mixed State:

- Suppose we prepare: State $|\psi_1\rangle$ with probability p_1
- State $|\psi_2\rangle$ with probability p_2
- This is not a superposition. It is a classical probability mixture.
- The the mixed state is: $\rho = p_1 |\psi_1\rangle\langle\psi_1| + p_2 |\psi_2\rangle\langle\psi_2|$

Mathematical Definition:

A density matrix ρ satisfies:

- Hermitian: $\rho^\dagger = \rho$
- Trace = 1 i.e. $Tr(\rho) = 1$
- Positive semidefinite: $\langle\phi|\rho|\phi\rangle \geq 0$
- Evolution: $\rho' = U\rho U^\dagger$

These three properties define all valid quantum states (pure and mixed)

Pure vs Mixed State Criterion

- Pure state: $\rho^2 = \rho$
- $Tr(\rho^2) = 1$ (Pure)
- $Tr(\rho^2) < 1$ (Mixed)
- Purity:
- Therefore, $$\mathrm{Tr}(\rho^2) = \frac{1 + |\vec{r}|^2}{2}$$
  - |r| = 1 → purity = 1 (pure)
  - |r| = 0 → purity = 1/2 (maximally mixed)

Note: The trace of a matrix is independent of the basis.

Density Matrix, Expectation Values

For pure states: $\langle A \rangle = \langle \psi \mid A \mid \psi \rangle$

- **The general formula of expectation value of an operator is:** $\langle A \rangle = Tr(\rho A)$
- **This works for both pure and mixed states. One unified formula for expectation values.**
- **Example: Single Qubit**
- **Pure State:** $\mid \psi \rangle = \alpha \mid 0 \rangle + \beta \mid 1 \rangle$

$$\rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}$$

- **Diagonal elements → probabilities**
- **Off-diagonal elements → coherences. These coherences encode quantum interference.**

For Completely Mixed State:

$$\rho = \frac{1}{2}I = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- **No coherence, maximum uncertainty**
- **Geometric Picture (Bloch Sphere)**
- **For one qubit: =** $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$ $\rho = \frac{1}{2}\begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}$
  - $\sigma = (\sigma_x, \sigma_y, \sigma_z)$
  - $\vec{r}$ **is Bloch vector**
  - $|\vec{r}| \leq 1$

The density matrix generalizes the state vector formalism to describe classical uncertainty, entanglement, and open-system dynamics in a single mathematically consistent framework.

Density Matrix: Takeaway

Why Density Matrices are important in Quantum Computing

- **They are essential for:**
  - **Modeling noise channels**
  - **Quantum process tomography**
  - **NISQ device analysis**
  - **Error correction theory**
  - **Lindblad master equations**
  - **Variational algorithms under noise**
  - **Thermal states**

In practice, hardware engineers need to think in density matrices.

Concepts covered so far…

- Kets → describe pure states
- Bras → dual vectors
- Operators → observables/unitaries (evolution)
- Outer product → projectors (measurement, noise)
- Density matrix → unified state formalism
- Classical mixture → statistical ensemble
- Partial trace → subsystem description

Coherence

Coherence

- **Hilbert space:** $H \in C^2$
- **The computational (z) basis:**
- $$|0\rangle = |+\rangle_z = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = |-\rangle_z = \begin{pmatrix}0\\1\end{pmatrix}$$
- **Observables are Pauli matrices:** $\sigma_x, \sigma_y, \sigma_z$
- **General pure spin state:** $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$
- **Define density operator:** $\rho = |\psi\rangle\langle\psi|$ =
$$\rho = \begin{pmatrix}|\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2\end{pmatrix}$$
- **The off-diagonal terms: are called as coherences**
$$\rho_{01} = \alpha\beta^*$$
- **The diagonal terms are the populations corresponding to the states**
- **Classical Mixture:**
$$\rho_{mix} = \begin{pmatrix}1/2 & 0 \\ 0 & 1/2\end{pmatrix}$$

If decoherence occurs (environment measures σ_z), off-diagonals decay:

$$\rho_{01}(t) = \rho_{01}(0)e^{-t/T_2}$$

Decoherence = decay of transverse spin components.

When coherence is present:

- Amplitudes interfere quantum advantage becomes possible
- When coherence is gone: only classical probabilities remain
- **Purity measures length of Bloch vector:**
$$P = \frac{1 + |\vec{r}|^2}{2}$$
- **Coherence measures projection onto the equatorial plane:**
$$x = \langle\sigma_x\rangle, \quad y = \langle\sigma_y\rangle, \quad z = \langle\sigma_z\rangle$$
$$C \sim \sqrt{x^2 + y^2}$$

A state is coherent in a given basis if its density matrix has non-zero off-diagonal elements in that basis.

Purity and Coherence

Coherence means "having phase relations between basis states."

- Consider the state: $|\psi\rangle = |0\rangle$
- The density matrix is:

$$\rho = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

- Fix a reference orthonormal basis $\{|i\rangle\}$. A state is coherent in a given basis if its density matrix has non-zero off-diagonal elements in that basis.
- In the computational basis: $\rho_{01} = 0$. Therefore, no coherence in the computational basis.
- But, if we check the purity: $Tr(\rho^2) = 1$
- Therefore, the state is pure but not coherent in the computational basis.
- But, the same state in x-basis:
- The density matrix in x basis does have off-diagonal elements and hence coherent.

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle_x + |-\rangle_x)$$

Purity is basis-independent. Coherence is basis-dependent.

Purity (Basis-Independent Property)

- Let ρ be a density operator on Hilbert space H.

ρ is pure $\Leftrightarrow \rho^2 = \rho$

- Equivalent condition: $Tr(\rho^2) = 1$
- If, $Tr(\rho^2) < 1 \Longrightarrow$ the state is mixed.
- Measure of purity: $\gamma(\rho) = Tr(\rho^2)$
- Purity is:
  - Basis independent
  - Invariant under unitary transformations
  - Global property of the state

Purity tells us how mixed the state is but not the origin of the mixedness.

Mixedness can come from: (i) Classical probability (ii) Entanglement with environment, (iii) Decoherence processes

It measures how much information about the system is inaccessible to us.

Entanglement

Entanglement

- **Let's consider two spin-1/2 systems:** $H_{Total} = C^2 \otimes C^2$
- The dimension of a 2-qubit system is: 4.
- The separable state is: $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$
- E.g. $|0\rangle \otimes |1\rangle$
- **Density Matrix factorizes as:** $\rho = \rho_A \otimes \rho_B$
- **The Bell State:** $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$
- **A bipartite pure state is entangled iff it cannot be written as:** $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$

Schmidt Decomposition

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |u_i\rangle \otimes |v_i\rangle$$

Any bipartite pure state:

$$\lambda_i > 0$$
$$\sum \lambda_i^2 = 1$$

With:

- If r=1 → separable
- If r>1 → entangled
- **For the singlet state:** $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$
- $\lambda_1 = \lambda_2 = 1/\sqrt{2}$
- Entanglement = global purity with local mixedness.
- Single-qubit coherence lives in off-diagonal terms
- **Entanglement lives in non-factorizability of the density matrix**

Qubit: Summary

Qubit: The Physical System

- A qubit is a two-level quantum system.
- **Examples:**
  - Spin-½ particle (electron, nucleus)
  - Polarization of a photon
  - Two-level atom
- **Mathematically, its state lives in a 2D complex Hilbert space:** $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

with normalization: $|\alpha|^2 + |\beta|^2 = 1$

Computational Basis:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

- These form an orthonormal basis.

Key Properties:

- Superposition
  - State exists in linear combination.
- Global Phase: Physically indistinguishable:

$$|\psi\rangle \sim e^{i\theta}|\psi\rangle$$

- **Bloch Sphere Representation**
- Multi-Qubit Systems: Tensor product structure: $|\psi\rangle \in C^2 \otimes C^2$
- E.g.: $$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
- This is entangled (cannot be factorized).

Thank you